Matching molecules to their NMR spectra with Trotterized time evolution on simulators and IBM Heron hardware

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The NMR spectrum follows from the correlation functions ($\alpha \in \{x, y\}$)

$$\mathrm{Tr}[\hat{I}_\alpha^{\mathrm{tot}}(t)\hat{I}_x^{\mathrm{tot}}(0)] = \sum_{n=1}^{2^N} m_n \langle m_n(t)|\hat{I}_\alpha^{\mathrm{tot}}|m_n(t)\rangle.$$

The task was to prepare all 2^N eigenstates $|m_n\rangle$ of $\hat{I}_x^{\mathrm{tot}}$, i.e. the X-basis product states $|\pm\rangle^{\otimes N}$. Listing 1 shows our solution. A Hadamard maps $|0\rangle \to |+\rangle$ and $R_z(\pi)$ maps $|+\rangle \to -i|-\rangle$, so $\phi_q \in \{0, \pi\}$ selects the X eigenvalue of qubit q (the global phase drops out).

I. MATCHING MOLECULES TO SPECTRA

We simulate each molecule with the notebook defaults (300 Trotter steps, $\Delta t = 2.5\,\mathrm{ms}$, window $e^{-\Gamma t}$ with $\Gamma = 5\,\mathrm{s}^{-1}$), interpolate the result onto the ppm grid of each reference and compute the Pearson correlation r (Table I). Looking at the total correlation gives A→3, B→2, C→1 (Fig. 1).

II. MAXIMUM TOLERABLE NOISE STRENGTH

For each molecule we increase the decay rate Γ of the window $e^{-\Gamma t}$ (log grid from 0.1–1000 s^{-1}, refined by bisection). Two neighboring lines count as resolved while the minimum between them stays below 99 % of the lower peak. $\Gamma_{\max}$ is the largest Γ at which all line pairs of the reference are still resolved; pairs closer than the frequency resolution $1/T \approx 1.3\,\mathrm{Hz}$ of the simulated signal (A: $J_{01} = 0.9\,\mathrm{Hz}$) can never be resolved and are excluded.

The window turns every line into a Lorentzian with FWHM Γ/π (in Hz), so $\Gamma_{\max}$ is set by the closest resolvable pair of lines. These are split by J-couplings ($J_{k\ell} = 2c_{k\ell}/\pi$, with $c_{k\ell}$ the coefficient of $X_k X_\ell$ in the Pauli Hamiltonian). For A and B the closest pair is the inner pair of a doublet of doublets, $\Delta\nu_{\min} = |J_a - J_b|$; for the two-spin molecule C it is $\Delta\nu_{\min} = J$ (Table II, Fig. 2).

TABLE I. Pearson correlation between simulated and reference spectra. Bold: optimal assignment. "–": no overlap in ppm.

	spectrum 1	spectrum 2	spectrum 3
molecule A	−0.04	−0.08	**0.69**
molecule B	–	**0.81**	−0.08
molecule C	**0.75**	−0.10	0.05

Listing 1. Our solution of the notebook task (notebook Sec. 5, initial states).

```
from qiskit.circuit import ParameterVector
def xsum_eigenstate_template(n_qubits: int):
    """Parameterized prep for any eigenstate of sum_i X_i."""
    phi = ParameterVector("phi", n_qubits)
    qc = QuantumCircuit(n_qubits, name="XsumEigen")
    # Each qubit: H takes |0> -> |+>, then Rz(phi) toggles
    ↪ between X eigenstates:
    #   phi = 0  -> |+>  (X eigenvalue +1)
    #   phi = pi -> |->  (X eigenvalue -1, up to a global phase)
    for q in range(n_qubits):
        qc.h(q)
        qc.rz(phi[q], q)
    return qc, phi
# Create the parametrized circuit.
qc_init_template, phi = xsum_eigenstate_template(number_spins)
# --- Enumerate initial states and build parameter maps ---
# All 2^N X-basis product states: bit 0 -> |+>, bit 1 -> |->.
bits_list = list(product((0, 1), repeat=number_spins))
# Bind phi[i] = pi * bit_i for each product state.
param_maps = [
    {phi[i]: np.pi * bits[i] for i in range(number_spins)} for
    ↪ bits in bits_list
]
```

TABLE II. Maximum tolerable decay rate (1 % dip). $\Delta\nu_{\min}$: smallest resolvable splitting and its origin. $\Gamma_{\max}$ in s^{-1}.

	$\Delta\nu_{\min}$ [Hz]	origin	$\Gamma_{\max}$
A	6.08	$J_{12} - J_{02}$	26.0
B	7.00	$J_{01} - J_{12}$	30.8
C	7.92	J_{01}	27.5

III. HARDWARE RUNS

a. Setup. We ran all three molecules on IBM Heron r2 processors (ibm_marrakesh, ibm_fez) with the Qiskit Runtime EstimatorV2: preset pass manager at optimization level 2 and 500 shots. All Trotter depths $k = 0 \ldots 59$ and all 2^N initial states (as parameter sets) were submitted as a single job per molecule. We compare with the noiseless statevector result and an exact density-matrix simulation with the FakeKingston noise model (Fig. 3, Table III).

b. Why 60 steps instead of 300. For the three-spin molecules one first-order Trotter step compiles to ≈ 13 CZ gates on the heavy-hex lattice (three Heisenberg pairs plus routing); the deepest of the 60 circuits already contains 795 CZ gates (depth ≈ 2800); 300 steps would require ≈ 4000 CZ gates. The cost also scales as

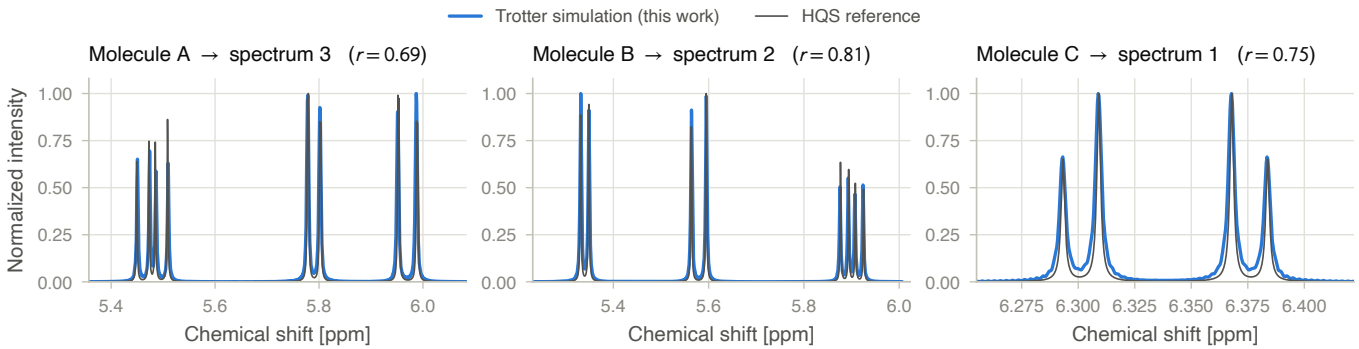


FIG. 1. Trotter-simulated spectra (blue; 300 steps of 2.5 ms, $\Gamma = 5\,\mathrm{s}^{-1}$) overlaid on the matched HQS reference spectra (gray). r: Pearson correlation of Table I.

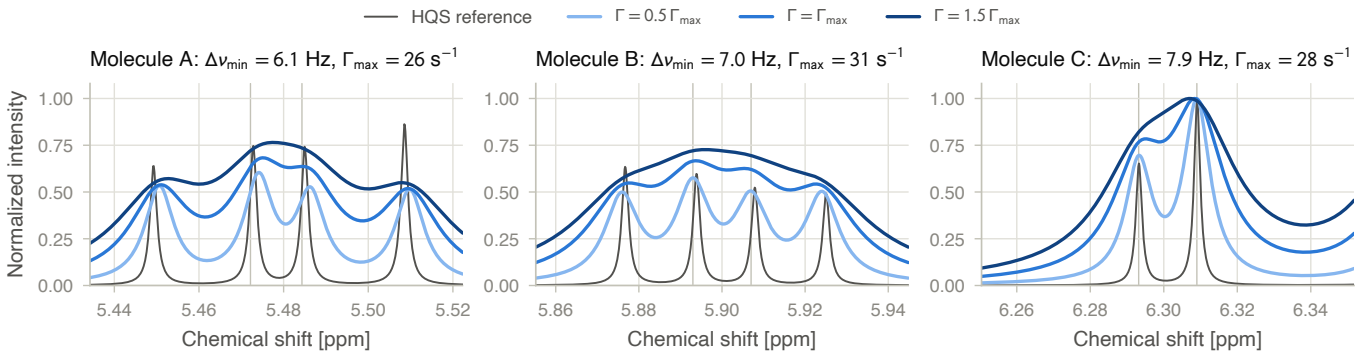


FIG. 2. Resolution limit. The multiplet containing the closest resolvable pair of lines (vertical rules) of each molecule at $\Gamma = 0.5,\ 1,\ 1.5\,\Gamma_{\max}$, with the HQS reference in gray. At $\Gamma_{\max}$ the closest pair still shows a 1 % dip; at $1.5\,\Gamma_{\max}$ it has merged.

$N_{\mathrm{steps}} \times 2^N \times 2$ bases×shots: 60 steps already used 177 s (A) and 167 s (B) of the 600 s monthly QPU budget.

c. Why a larger time step. At a fixed number of steps, i.e. fixed circuit depth, a larger time step covers a longer evolution time and therefore gives a finer frequency resolution. Two effects bound it from above: the signal must be sampled fast enough to capture the outermost lines without aliasing, and the Trotter error, which grows with the step size, must not shift the lines noticeably. For A and B, whose lines span a wide frequency range, the sampling condition sets the step. The narrow spectrum of C allows a considerably larger step; moreover, for two qubits the transpiler compresses any number of Trotter steps into a single two-qubit unitary, so the circuit depth of C does not grow with time.

TABLE III. Hardware runs (500 shots). CZ: two-qubit gates in the deepest circuit; QPU: billed time.

	backend	Δt [ms]	T [s]	CZ	QPU [s]
A	ibm_marrakesh	3.0	0.18	795	177
B	ibm_fez	3.0	0.18	795	167
C	ibm_marrakesh	7.5	0.45	3	76

d. Results. All lines of all three molecules appear at the correct positions on hardware (Fig. 3). For C, hardware, noise model and ideal result are indistinguishable. For A and B the hardware deviates more from the ideal signal than the FakeKingston model predicts, but all multiplets are reproduced. The resolution of the A and B hardware spectra is limited by the short record ($1/T = 5.6$ Hz), which is in turn set by circuit depth; shot noise and gate errors appear as a noisy baseline between the multiplets.

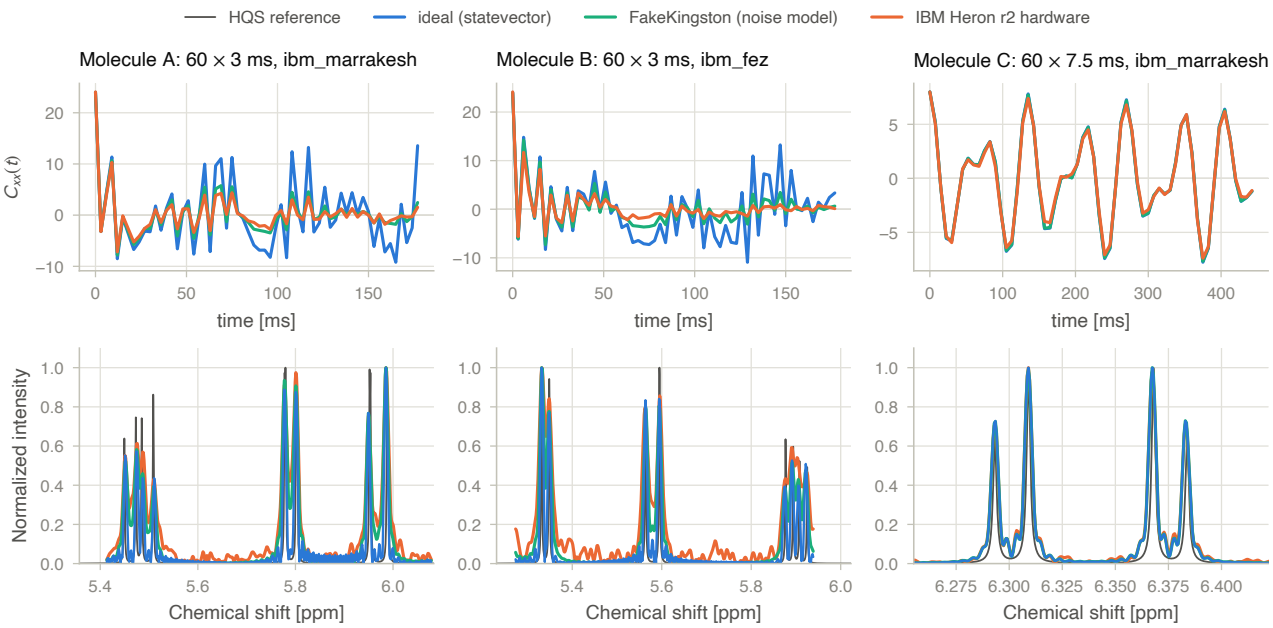


FIG. 3. Hardware results. Top: correlation function $C_{xx}(t)$ (no window) from the ideal statevector simulation, the FakeKingston noise model and IBM Heron r2 hardware. Bottom: corresponding spectra ($\Gamma = 5\,\mathrm{s}^{-1}$, 8-fold zero padding) with the HQS reference. For the two-spin molecule C every Trotter depth compiles to ≤ 3 CZ gates, so the hardware matches the ideal result.